Annamacharya Institute of Technology and Sciences (Autonomous)
Venkatapuram (V), Karakambadi Road, Renigunta (M), Tirupati.
Department of Computer Science Engineering

Data Structures

Questions – 2Marks

UNIT - 1

1. Define a linear data structure.
2. List any two examples of linear data structures.
3. What is an Abstract Data Type (ADT)?
4. Write two differences between ADT and Data Structure.
5. What is the importance of linear data structures in programming?
6. Define time complexity.
7. Define space complexity.
8. What is the time complexity of traversing a linear data structure of size n?
9. Define Linear Search.
10. Write the time complexity of Linear Search in the worst case.
11. Define Binary Search.
12. What is the prerequisite condition to apply Binary Search?
13. Write the time complexity of Binary Search.
14. Define Sorting.
15. What is Bubble Sort?
16. Write the worst-case time complexity of Bubble Sort.
17. What is Selection Sort?
18. Write the main idea behind Selection Sort.
19. What is Insertion Sort?
20. Write the best-case time complexity of Insertion Sort.

UNIT - 2

1. Define a Linked List.
2. What is a node in a linked list with a sketch?
3. Write the structure of a singly linked list node in C with a sketch.
4. What are the components of a node in a singly linked list with a sketch?
5. Define a head pointer in a linked list.
6. Write the algorithm steps to insert a node at the beginning of a singly linked list.
7. What is the time complexity of inserting a node at the beginning of a singly linked list?
8. What is the time complexity of deleting the last node in a singly linked list?
9. Write a C statement to traverse a singly linked list using a pointer temp.
10. What happens if the head pointer is NULL?
11. Define a Doubly Linked List.
12. Write the structure of a doubly linked list node with a sketch.
13. Mention two advantages of a doubly linked list over a singly linked list.
14. How many pointers are present in a doubly linked list node with a sketch?
15. Define a Circular Linked List.
16. What is the main difference between a singly linked list and a circular linked list?
17. How do you identify the end of a circular linked list during traversal?
18. Write any two differences between Arrays and Linked Lists.
19. Mention two applications of linked lists.
20. Write a small code snippet to count the number of nodes in a singly linked list.

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UNIT – 3

1. Define a stack.
2. What is the LIFO principle?
3. List any two applications of stacks.
4. What are the basic operations performed on a stack?
5. Define the PUSH operation.
6. Define the POP operation.
7. What is the PEEK operation in the stack?
8. What is stack overflow?
9. What is stack underflow?
10. How is a stack implemented using arrays?
11. How is a stack implemented using linked lists?
12. What is the advantage of the linked list implementation of a stack?
13. Write the condition for stack overflow in the array implementation.
14. Write the condition for stack underflow.
15. What is the role of the TOP pointer in stacks?
16. How are stacks used in recursion?
17. How are stacks used in expression evaluation?
18. What is a postfix expression?
19. What is an infix expression?
20. Mention the application of stacks in backtracking.

UNIT – 4

1. Define a queue.
2. What is the FIFO principle?
3. List the basic operations of a queue.
4. Define the ENQUEUE operation.
5. Define the DEQUEUE operation.
6. What is queue overflow?
7. What is a queue underflow?
8. How is a queue implemented using arrays?
9. How is a queue implemented using linked lists?
10. What is a circular queue?
11. What is the advantage of a circular queue over a linear queue?
12. Define deque.
13. What are the types of deque?
14. Differentiate between a stack and a queue.
15. What is the role of FRONT and REAR in queues?
16. Mention any two applications of queues.
17. How are queues used in scheduling?
18. What is Breadth First Search (BFS)?
19. How are queues used in BFS traversal?
20. What is a priority queue?

UNIT – 5

1. Define a tree data structure.
2. What is a node in a tree?
3. Define root node.
4. Define a leaf node.
5. What is a Binary Search Tree (BST)?
6. What are the properties of BST?
7. Define inorder traversal.
8. Define preorder traversal.
9. Define postorder traversal.
10. What is tree traversal?
11. What is insertion in a BST?
12. What is deletion in BST?
13. Mention any two applications of trees.
14. Define hashing.
15. What is a hash table?
16. Define hash function.
17. What is a collision in hashing?
18. What is chaining in hashing?
19. What is open addressing?
20. Mention any two applications of hashing.